Experiment - 2

Network Cables

AIM:

To study different types of network cables and to construct straight-through and crossover Ethernet cables using a crimping tool.

PROCEDURE:

1: Identify different network cables:

→ UTP

→ STP

→ Coaxial

→ Fibre optic

2. Take a CAT5/CAT5e ethernet cable, RJ45 connectors and a crimping tool.
3. Strip approximately 1.5 cm of the cable covering.
4. Untwist and arrange the wires according to the required wiring standard.
5. For a straight-through cable, use the same arrangement at both ends.
6. For a crossover cable, use arrangement A at one end and B at the other.
7. Insert the wires into the RJ45 connector.
8. Place the connector inside the crimping tool & crimp it.
9. Repeat the process for the other end.
10. Test the cable by connecting the appropriate network devices.

The manual identifies UTP, STP, coaxial and fibre-Optic cables and gives the crimping procedure, including stripping, arranging, inserting and crimping the wires.

Sample Output:

Straight-through cable:
End A → T568B
End B → T568B

Crossover Cable:
End A → T568A
End B → T568B

Cable Tested Successfully.

Result:

Thus, different types of network cables were studied and straight-through & crossover Ethernet cables were Successfully constructed and tested.